



PETRONAS

SKG 12 Mechanical Engineering

Predictive Maintenance Technique – Part 1 (Failure Mechanism), L2

Failure Mechanisms and Detection Parameters (3 of 4)

Rotating Equipment

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Failure Mechanisms and Detection Parameters for Rotating Equipment

Learning Objectives

By the end of this session, you should be able to:

- Understand the various degradation mechanisms affecting equipment components i.e. RCFA approach
- Understand the failure physics relating the degradation parameters for specific failure mechanism i.e. the deterministic approach, to be used in RCFA and detecting incipient failures through Predictive Maintenance Techniques.
- Aware of the re-engineering and RCFA approach that can be implemented to design out or avoid recurrence of failure.
- Aware to the various technology for detection of degradation parameters, and interpretation of results.

Failure Mechanisms related following Rotating Machinery Covered

- Centrifugal Pump
- Centrifugal Compressors
- Reciprocating Compressors
- Combustion Engine
- Gas Turbine
- Steam Turbine

Pre-requisite Information

Technical drawings/diagrams containing process data, material and media codes:

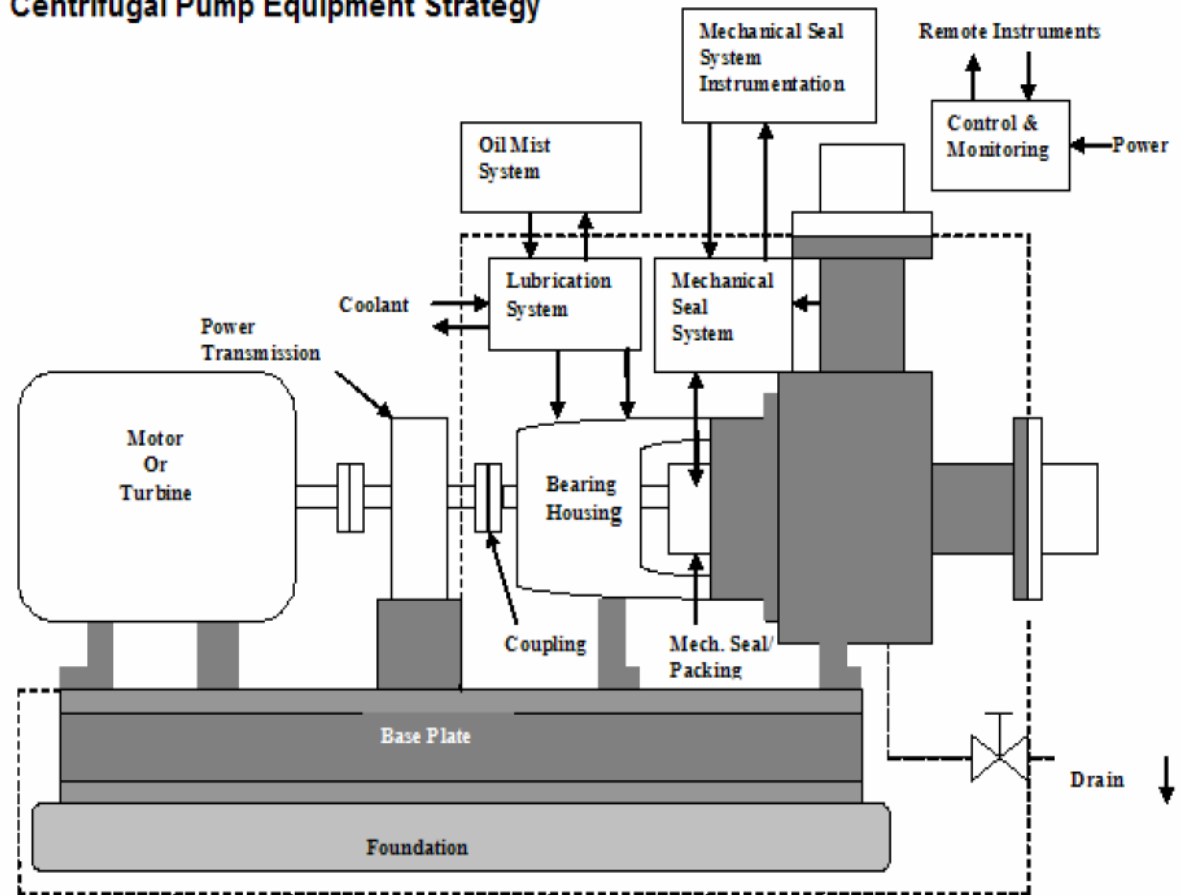
- PFD & P&IDs
- Shut down logic if any
- Cause and effect diagrams if applicable
- Fire protection data sheets if applicable
- Asset Structure / Acceptance Criteria

Centrifugal Pump

Pump Failure Mechanisms

- Mechanical Seal Wear
- Bearing Wear Out
- Lubrication System Failure
- Cooling System Failure
- Pump / Driver Misalignment
- Disk Pack Fracture
- Fouling
- Erosion of Impeller or Casing
- Outside Operating Envelope
- Plugged Suction
- Excessive Starts / Swaps
- Blocked Discharge

Centrifugal Pump Equipment Strategy



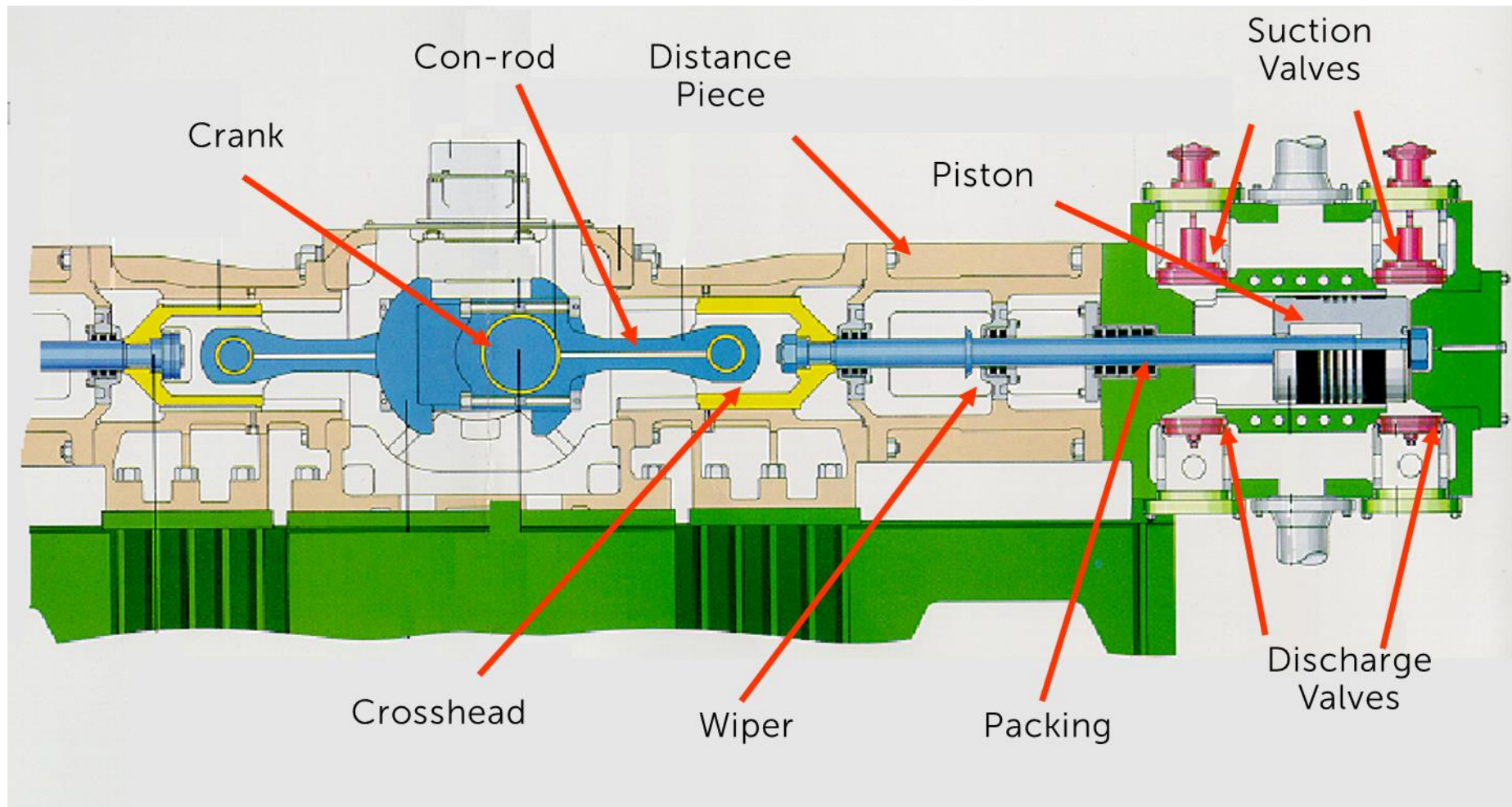
Centrifugal Pump

Failure Mechanism	Failure Mode	Detection Parameter
Mechanical Seal Wear	Leakage above tolerable limit (< 4dips / minute)	1. Leakage Rate 2. Critical Instrumentation (Deviations)
	Gross Seal Leakage (LOPC)	
Bearing Wear Out	Noise, excessive vibration or heat.	1. Vibration measurement 2. Noise measurement 3. Lube Oil Analysis measurement
	Loss of shaft support, rubbing or seizure of rotating components, possible shaft fracture	
Lubrication system failure	Noise, excessive vibration or heat.	1. Bearing Temperature 2. Noise and Vibration 3. Lube Oil Analysis measurement
	Loss of shaft support, rubbing or seizure of rotating components, possible shaft fracture	
Cooling system failure	Loss of shaft support, rubbing or seizure of rotating components, possible shaft fracture	1. Bearing Temperature 2. Vibration measurement 3. Operating Parameter
	Bearing Overheat	
Pump/Driver Misalignment	High vibration shutdown	1. Vibration measurement 2. Bearing Temperature
	Bearing failure, loss of shaft support, rubbing of rotating components, shaft fracture.	

Centrifugal Pump

Failure Mechanism	Failure Mode	Detection Parameter
Disk pack fracture	High vibration shutdown / No pump capacity	1. Vibration measurement 2. Noise measurement 3. Operating Parameter
	Bearing failure, loss of shaft support, rubbing of rotating components, shaft fracture.	
	Coupling components ejected at high Velocity	
Fouling	Loss of capacity	1. Operating Parameters 2. Vibration measurement
	High vibration shutdown	
Wear / erosion of impeller or casing	Loss of capacity	1. Operating Parameters 2. Vibration measurement
	High vibration shutdown	
Operation outside Envelop	Loss of capacity	1. Operating Parameters 2. Vibration measurement
	High vibration	
	Loss of containment	
Plugged suction	Vibration	1. Operating Parameters 2. Vibration measurement
	Loss of containment	
Excessive starts / swaps	Mechanical Seal / Bearing Failure (refer respective	1. Leakage Rate 2. Critical Instrumentation (Deviations)
Blocked Discharge	Vibration / Loss of containment	1. Vibration measurement 2. Bearing Temperature

Reciprocating Compressor



Reciprocating Compressor

Sno	Failure Mechanism	Detection Parameter
01	Liquid / Solids Ingestion	1. Operating Parameters 2. Vibration measurement
02	Operation Outside Design Envelope	1. Operating Parameters 2. Vibration measurement
03	Valve Failure	1. Valve Temp measurement 2. Discharge Pressure
04	Packing Failure	1. Temp profiling 2. Discharge Pressure
05	Frame Lubrication System	1. Temp profiling 2. Noise measurement
06	Cylinder / Packing Lubrication System	1. Temp profiling 2. Noise measurement
07	Fatigue of Machine Bolting	1. Vibration measurement 2. Noise measurement
08	Machine Bearing Failure	1. Vibration measurement 2. Noise measurement 3. Operating Parameter
09	Flexible Coupling Failure	1. Vibration measurement 2. Noise measurement 3. Operating Envelop (PV Curve)

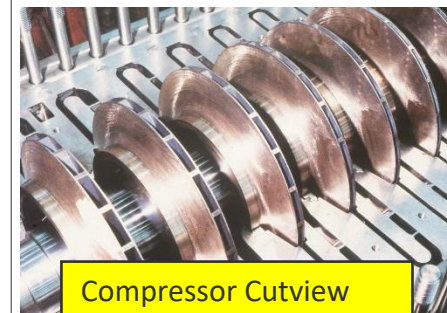
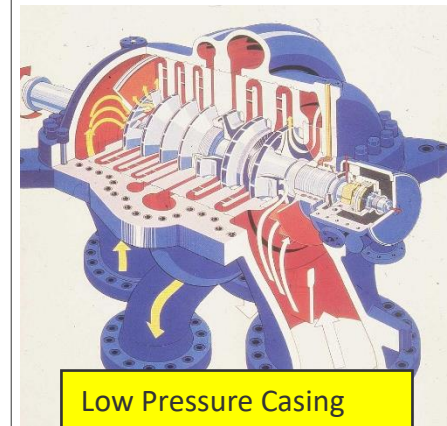
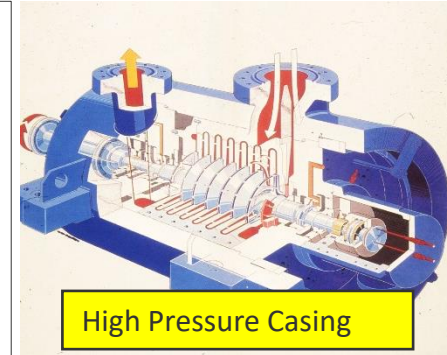
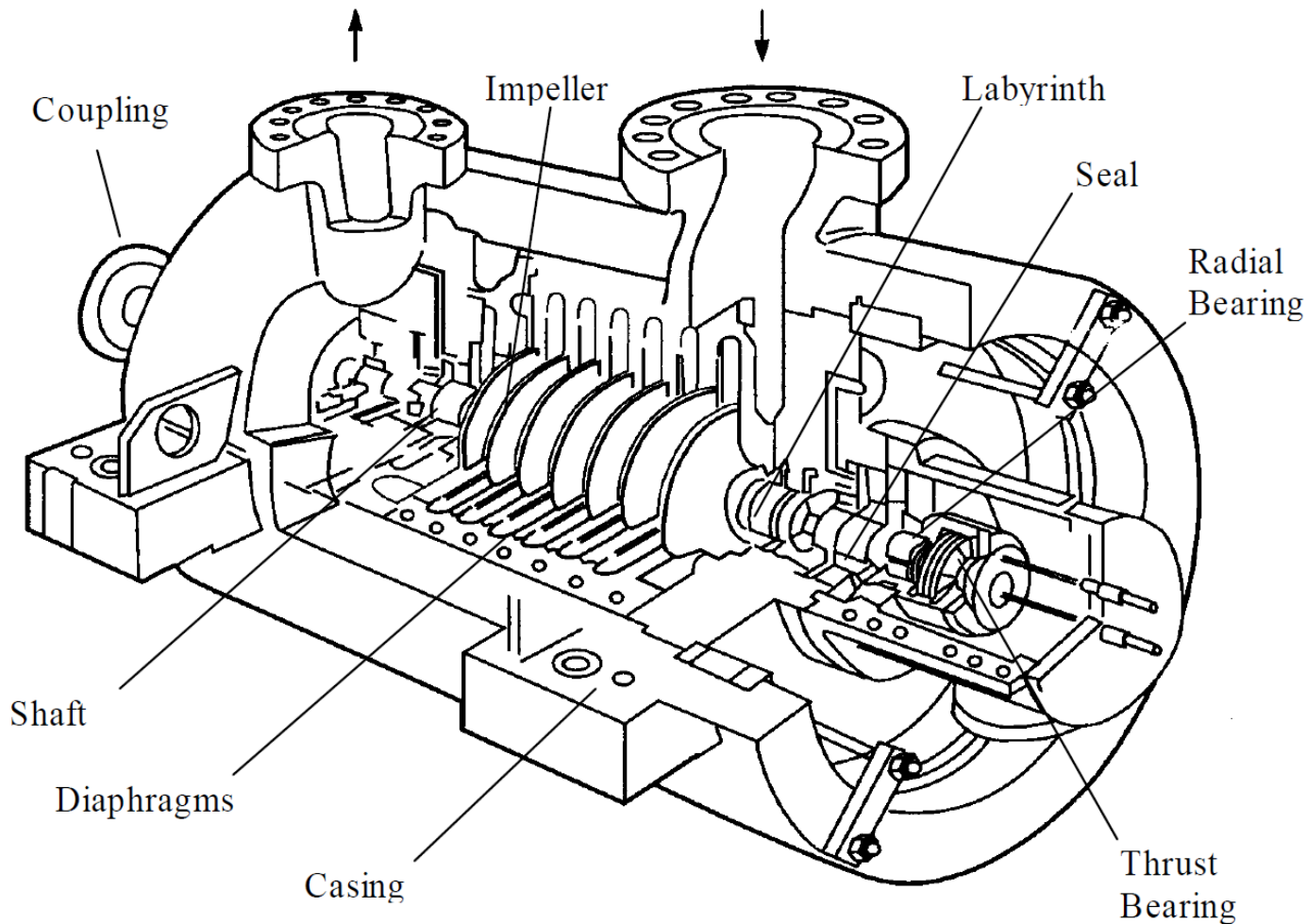
Reciprocating Compressor

Sno	Failure Mechanism	Detection Parameter
11	Acoustic Pulsation and Resonance	1. Vibration measurement 2. Noise measurement
12	Crankshaft Fatigue	1. Vibration measurement 2. Noise measurement 3. Operating Envelop (PV Curve)
13	Piston Failure	1. Rod drop / displacement 2. Noise measurement 3. Operating Envelop (PV Curve)
14	Piston Rod Failure	1. Rod drop / displacement 2. Vibration measurement 3. Operating Envelop (PV Curve)
15	Piston Rod / Crosshead Nut Loosening	1. Rod drop / displacement 2. Noise measurement 3. Vibration measurement
16	Cylinder Liner, Ring Wear Out	1. Rod drop / displacement 2. Operating Envelop (PV Curve)
17	Compressor Cooling System Failure	1. Bearing Temp measurement 2. Vibration measurement 3. Operating Envelop (PV Curve)

Centrifugal Compressor

Centrifugal Compressor Cut-View with internals, Casing Designs and Impeller

SPECIAL PURPOSE CENTRIFUGAL COMPRESSOR



Centrifugal Compressor

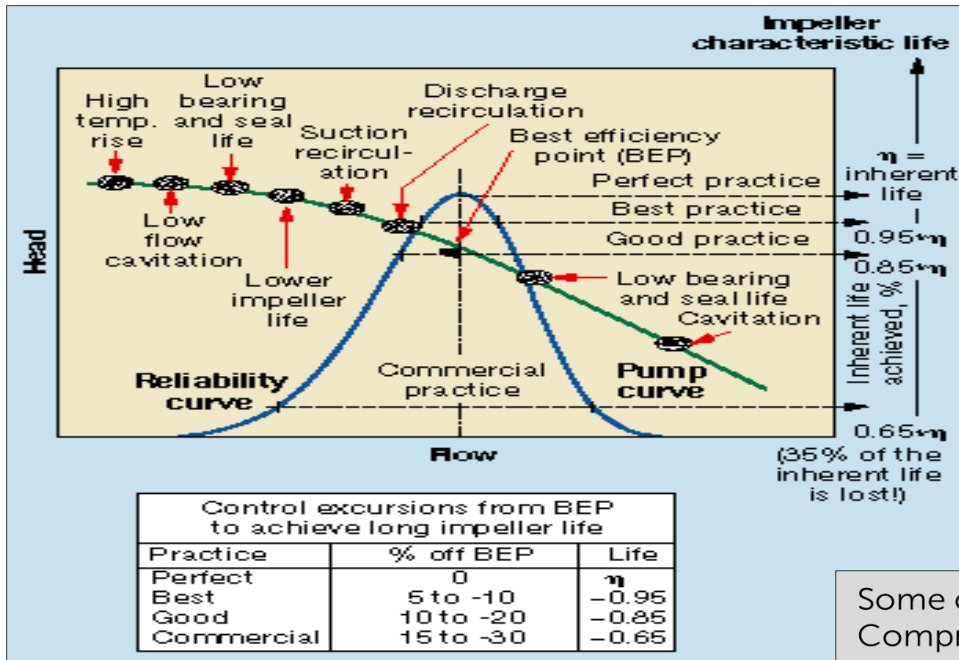
Component	Failure Mechanism	Detection Parameter
Rotor	Liquid Slugging	1. Vibration measurement 2. Operating Parameter
Complete Assembly	Extended Compressor Surging	1. Vibration measurement 2. Noise measurement 3. Operating Parameter
Complete Assembly	Operation outside Design	1. Vibration measurement 2. Operating Parameter
Rotor	Internal Fouling	1. Vibration measurement 2. Operating Parameter
Rotor	Internal Corrosion / Erosion	1. Vibration measurement 2. Operating Parameter
Lubrication	Degradation of Lube Oil	1. Bearing Temperature measurement 2. Noise and Vibration measurement 3. LO Properties
Radial Bearing	Radial Bearing Vibration	1. Vibration measurement 2. Noise measurement 3. LO Temp Measurement
Thrust Bearing	Thrust Bearing Vibration	1. Thrust measurement 2. Noise measurement 3. LO Temp Measurement

Centrifugal Compressor

Component	Failure Mechanism	Detection Parameter
Seal Oil	Seal Oil Degradation	1. Seal Leak rate / Pressure 2. Operating Parameter
Oil Seal	Oil Seal Failure	1. Oil Seal Leak rate 2. Operating Parameter
Gas Seal	Gas Seal Failure	1. Gas Seal Leak rate / Pressure 2. Operating Parameter
Rotor	Rotor High Vibration	1. Vibration measurement 2. Operating Parameter
Impeller	Impeller Failure	1. Vibration measurement 2. Operating Parameter
Shaft	Shaft Fatigue	1. Vibration measurement 2. Operating Parameter
Lubricated Coupling	Lubricated Coupling Failure	1. Vibration measurement 2. Operating Parameter 3. LO Temp measurement
Non Lube Coupling	Non-Lubricated Coupling Failure	1. Vibration measurement 2. Operating Parameter
Gaskets	External / Internal Material Degradation	1. Leak rate / 2. Product Quality
Rotor	Foreign Object Damage	1. Vibration / Noise Measurement 2. Operating Parameter

Centrifugal Pump / Compressor

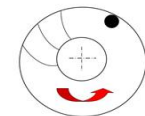
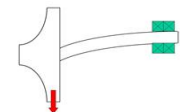
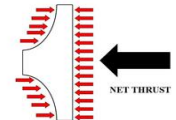
Failure Mechanism Explained



Flow Vs Head Hydraulic Characteristic Curve for typical Centrifugal Pump. Curve depicts Failure Mode / Mechanisms associated at different flow and Head patterns.

Some of the forces inside Centrifugal Pump / Compressor:

- Axial thrust
- Radial thrust
- Weight of rotating element (impeller/shaft)
- Imbalance (rotating element)



Some of internal forces or Failure Mechanisms acted on various components of Centrifugal Pump or Centrifugal Compressor which eventually lead to failures.

Centrifugal Pump / Compressor

Failure Mechanism – Bearing failure

Fatigue

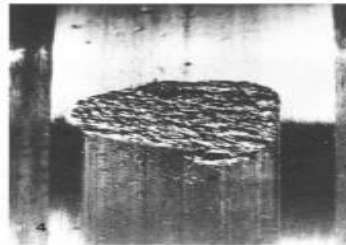
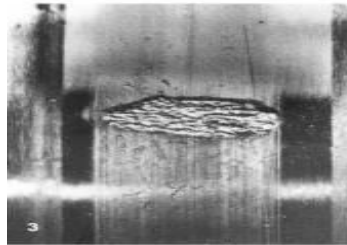
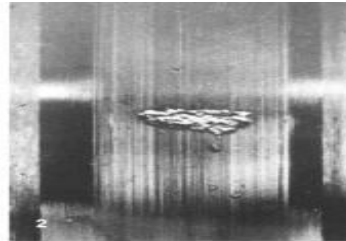
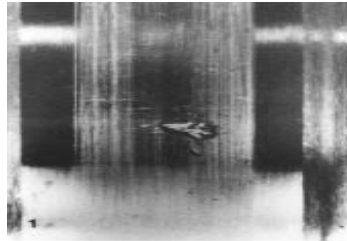
Wear

Corrosion

Electrical erosion

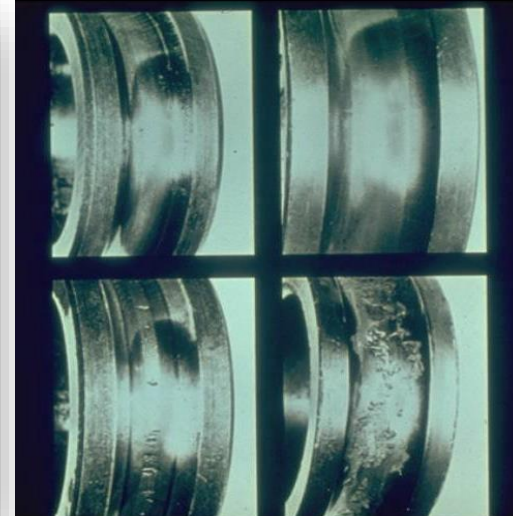
Plastic deformation

Fracture



Under defined operating conditions, Fatigue is considered as normal end to useful life. However Fatigue is accelerated by Repeated stress or load changes, Improper mounting / loading, Improper Lubrication. It can be detected by Periodic Vibration and Noise Analysis.

Fatigue is caused by a fixed load acting on a metallic body at a periodic interval (Accelerating by improper loading, lubrication & metallurgical defect). Starting with metallurgical distortion at a depth of 200 – 500 Micron below surface with final manifestation as Spalling.

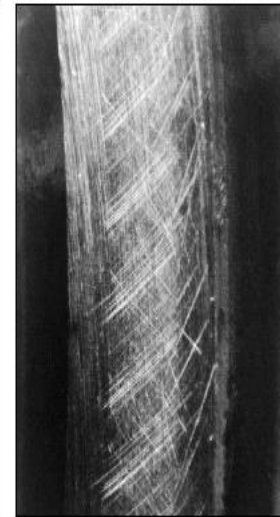
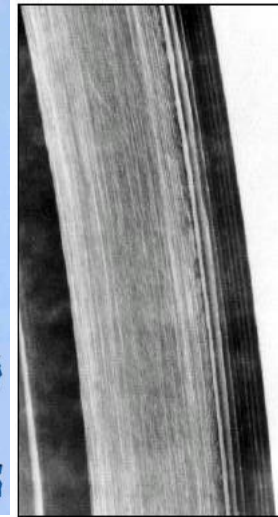
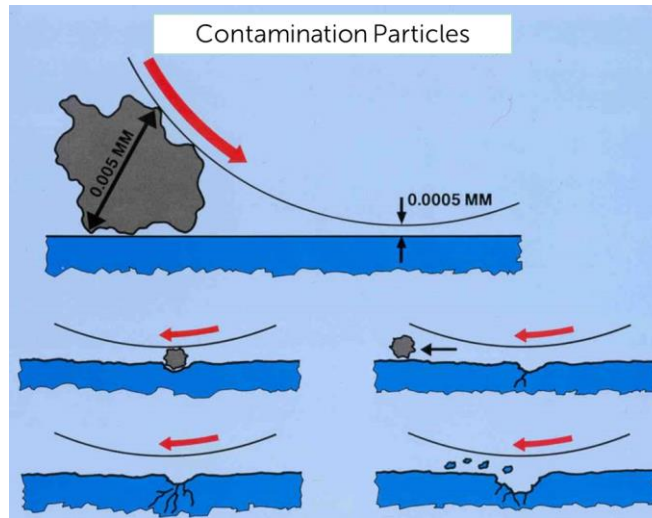


Centrifugal Pump / Compressor

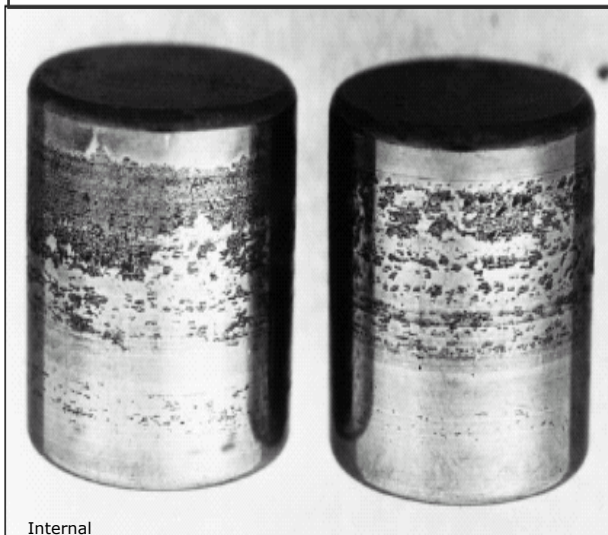
Failure Mechanism – Bearing failure

Wear

Wear is Progressive Removal of Material and accelerating with Inadequate Lubrication, Ingress of Dirt Particles. Wear can be detected by periodic Lube Oil, Vibration & Bearing Temperature Analysis.



Ingress of dirt particle as small as 5Micron could initiate rapid wear and ultimately fracture. Also bearing wear is caused by improper mounting, inadequate lubrication (Temp over 125°C results in loss of hardness) and Improper Lubrication(different grades of viscosity).

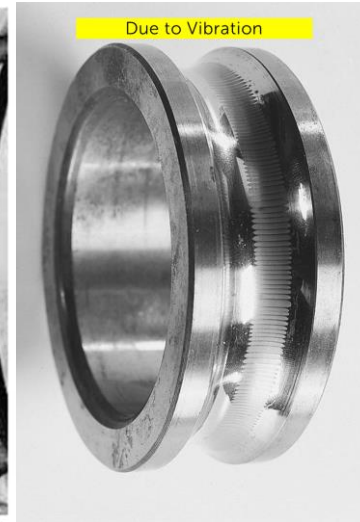
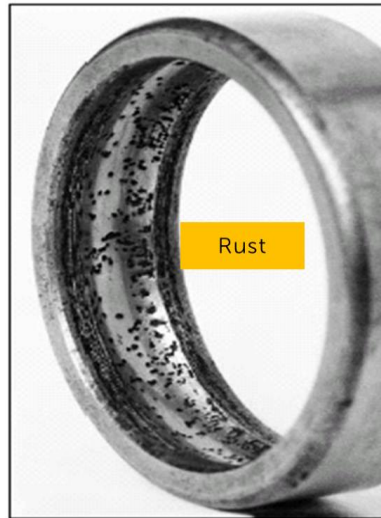


Centrifugal Pump / Compressor

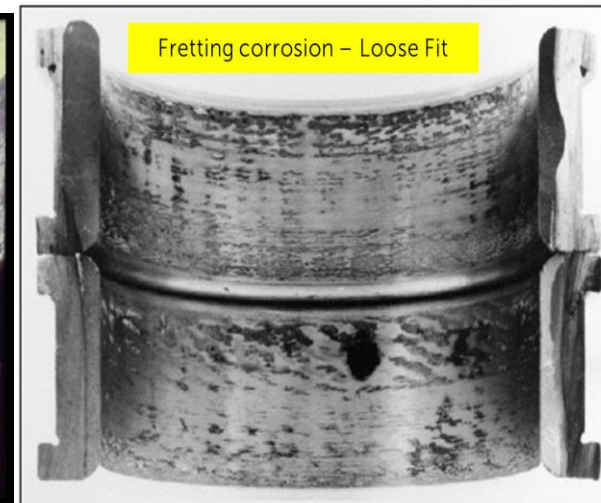
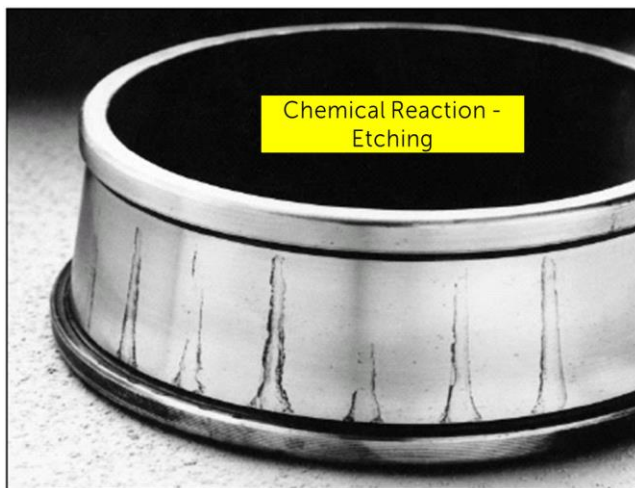
Failure Mechanism – Bearing failure

Corrosion

Corrosion in bearing is caused by Oxidation, rust or Chemical reaction due to Frictional force or Moisture presence in Lube Oil (oil water mixture). It can be seen as Pits, flaking.



Bearing failure caused by Corrosion of the material is predominantly Moisture in LO or Frictional force generated due to improper fitting, vibration, looseness. Corrosion can be reduced by periodic Lube Oil analysis. However, once affected by corrosion, the bearing requires maintenance or replacement.

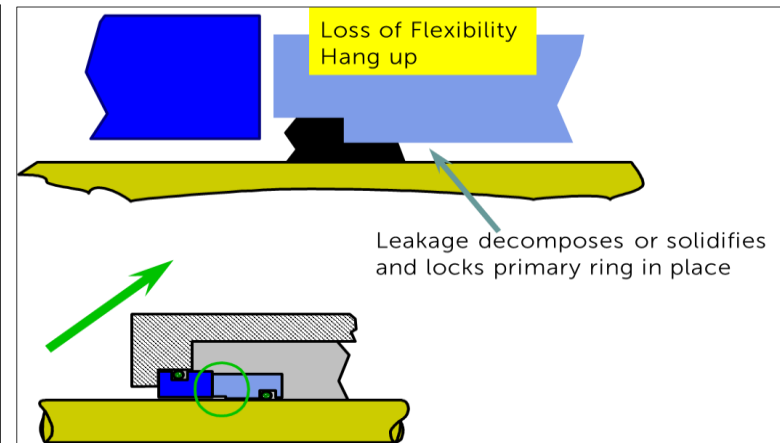
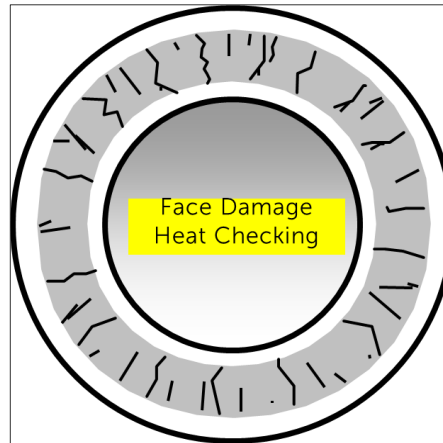


Centrifugal Pump / Compressor

Failure Mechanism – Mechanical Seal Failure

Failure Mechanism

Failures are due to incorrect
Installation
Machinery design and operation
Seal design and operation
Affected component / Zone
Seal Faces
Loss of Flexibility or Hang Up
Secondary Sealing Elements
Material Compatibility.



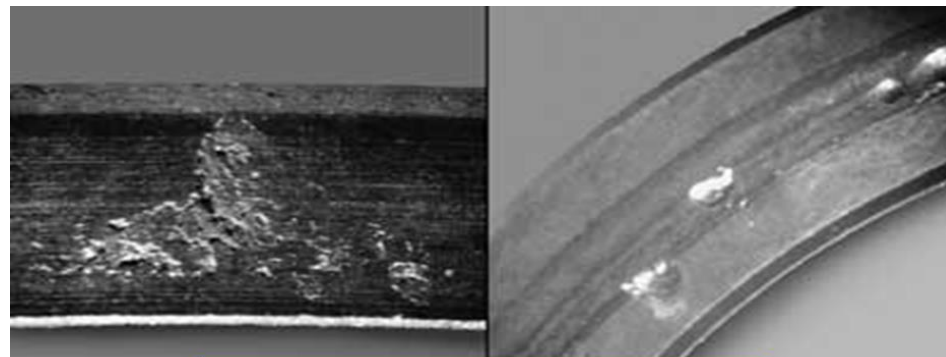
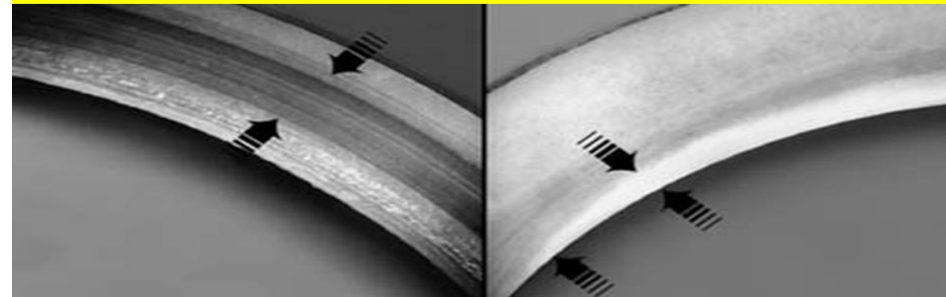
SCORING OR EROSION

SEAL FACE SCORING

SEAL FACE EROSION

Seal face incompatibility, Exceeded limits (speed, temperature, pressure), Misapplication, Improper installation, Improper seal face lubrication or flushing are some of FM resulted in failures. Basic detection parameters are Leak rate, Loss of Level or Pressure in Seal Pot(double seal).

STATIONARY FACE - CENTERED ROTARY FACE- MISCENTERED



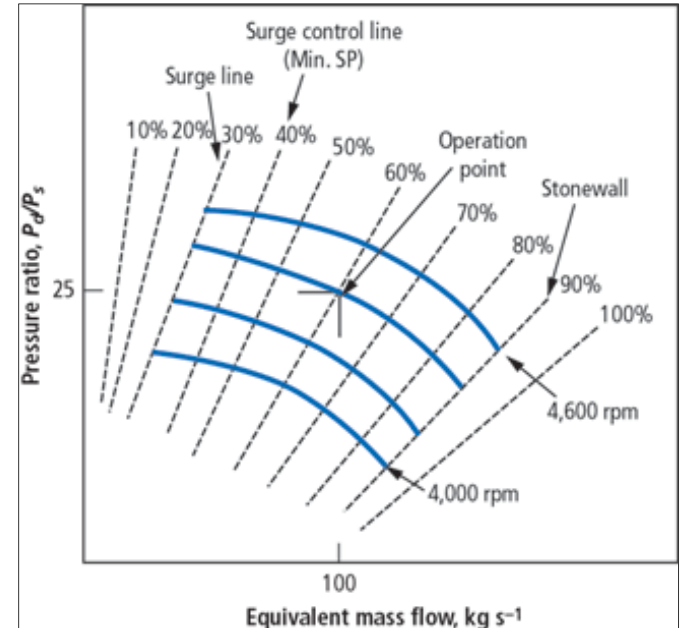
BLISTERING OF SEAL FACES

Centrifugal Compressor

Failure Mechanism – Surge

Antisurge control:

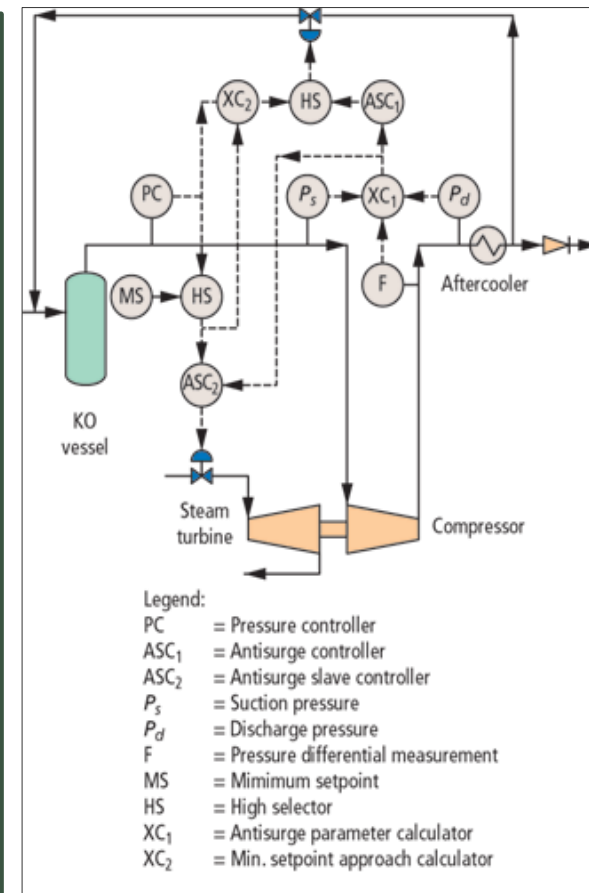
- To predict when a compressor is entering the surge region, an invariant compressor curve is required. A compressor curve is called invariant when the curve shape and height from the axis doesn't change with changes in Mol-weight, temperature, pressure and compressibility.
- The polytropic head versus actual volume flow is an invariant compressor curve since it is independent of changes in suction pressure, and its shape is only slightly influenced by gas composition and suction temperature.
- Extrapolation of this antisurge parameter over the compressor's operating window of each compressor stage allows unambiguous determination of the compressor's operating point in its operating window.



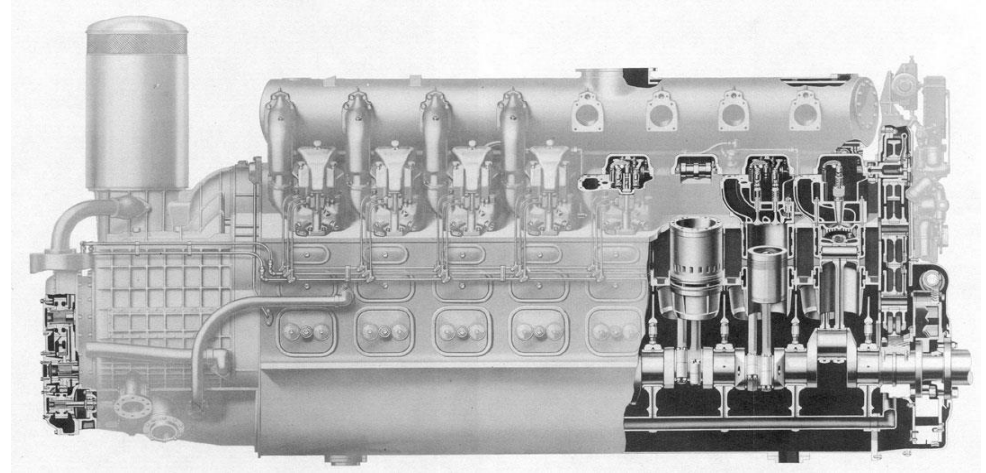
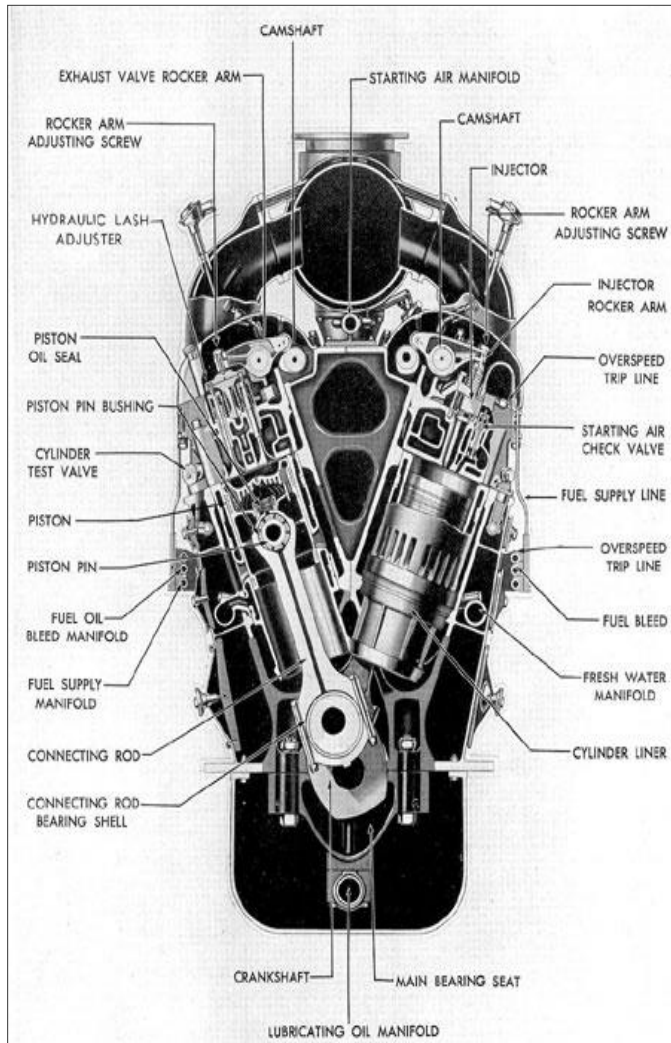
Centrifugal Compressor

Failure Mechanism – Surge

1. Verify Compressor has Anti-Surge Control system.
2. Verify an “approaching surged or minimum flow alarm exists in the dedicated compressor control system or process control system to allow compressor operators to adjust operating parameters to avoid potential surge conditions. The above low flow pre-alarms are used where operations prefer to be alerted prior to operation of the anti-surge protection system which may cause sudden decrease in compressor capacity and possible process upset.
3. Verify compressor operating envelope is adequately defined with respect of surge and that operators are trained how to respond to low flow conditions.
4. Verify anti-surge systems are designed to protect compressor from surge during all startup and shutdown sequences.
5. Verify a suitable surveillance, monitoring, preventive and predictive maintenance program for the Anti-Surge Control System critical instruments is in effect or establish one if it is not.



Combustion Engine



Combustion Engine

Component	Failure Mechanism	Detection Parameter
Fuel	Fuel Contamination	1. Engine Performance 2. Fuel Sample Analysis
Fuel	Loss of Fuel	1. Engine Performance
Fuel Filter	Plugged Filter or Damaged Filter	1. Engine Performance 2. Delta Pressure Across Filter
Fuel Pump	Low Pressure	
Injector Spray Nozzle	Plugged Spray Nozzle	
Injector Spray Nozzle	Erosion / Corrosion	
Inlet/Exhaust Valves	Sticky Valve Stem	
Inlet/Exhaust Valves	Valve Leak	
Inlet/Exhaust Valves	Valve Timing Faulty	
Starting Air System	Low Pressure	1. Engine Performance 2. RPM
Lubrication System	Degradation of Lube Oil	1. Bearing Temp measurement 2. Noise / Vibration measurement 3. LO Properties

Combustion Engine

Component	Failure Mechanism	Detection Parameter
Lubrication System	Lubrication System Failure	1. Engine Performance 2. Bearing Temperature
Cylinder	Wear	1. Rod drop / displacement 2. Noise measurement 3. Operating Envelop (PV Curve)
Piston and Rings	Wear	1. Rod drop / displacement 2. Noise measurement 3. Operating Envelop (PV Curve)
Governor	Governor Hunting	1. Engine Performance 2. RPM
Cooling System	Cooling System Failure	1. Bearing Temperature measurement 2. Vibration measurement 3. Operating Parameter
Bearings	Wear	1. Vibration measurement 2. Noise measurement 3. LO Temp Measurement
Whole	Operating Outside Design Envelope	1. Vibration measurement 2. Operating Parameter
Cylinder / Piston	Foreign object damage	1. Vibration / Noise Measurement 2. Operating Parameter
Coupling	Misalignment	1. Vibration / Noise Measurement 2. Operating Parameter

Gas Turbine



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Thank you